

Given by frequency-response tuning

P controller

$$K_c = 0.7, \quad T_c = 0.2566$$

$$K_P = 0.5K_c = 0.35$$

PI controller

$$K_P = 0.4K_c = 0.28, \quad T_I = 0.8T_c \approx 0.21$$

PID controller

$$K_P = 0.6K_c = 0.42, \quad T_I = 0.5T_c \approx 0.13, \quad T_D = 0.125T_c \approx 0.03$$

message

when the closed loop is unstable, frequency-response tuning is still usable even though the step-response method is not